

IVP: $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$, $D: |x - x_0| < a$, $|y - y_0| < b$.
— (1)

Integrate w.r to x from x_0 to x ,

$$\int_{x_0}^x dy = \int_{x_0}^x f(t, y(t)) dt$$

$$\Rightarrow y(x) - y(x_0) = \int_{x_0}^x f(t, y(t)) dt$$

$$\Rightarrow \boxed{y(x) = y(x_0) + \int_{x_0}^x f(t, y(t)) dt} \quad (2)$$

The Eq. (2) is called the integral equation in y .

Picard's iterations: Start with $y_0(x) = y_0$ and successively define

$$y_{n+1}(x) = y_0 + \int_{x_0}^x f(t, y_n(t)) dt, \text{ then}$$

$\lim_{n \rightarrow \infty} y_n(x) = y(x)$, where $y(x)$ is the solution of IVP (1).

Ex: $y' = 1 + xy$, $y(0) = 1$.

Sol: $y_0(x) = y_0 = 1$.

$$y_1 = y_0 + \int_0^x f(t, y_0(t)) dt$$

$$= 1 + \int_0^x f(t, 1) dt = 1 + \int_0^x (1 + t) dt$$

$$= 1 + \left(t + \frac{t^2}{2} \right) \Big|_0^x = 1 + x + \frac{x^2}{2}.$$

$$y_2 = y_0 + \int_0^x f(t, y_1(t)) dt = 1 + \int_0^x (1 + ty_1) dt$$

$$= 1 + \int_0^x \left(1 + t \left(1 + t + \frac{t^2}{2} \right) \right) dt$$

$$= 1 + \left[t + \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{8} \right]_0^x$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8}$$

$$y_3 = ? \quad , \quad y_4 = ?$$

Ex. Find the 4th iteration of the following diff eq.
using Picard's iteration.

(i) $y' = 2\sqrt{y}$, $y(1) = 0$ (ii) $y' = x - y^2$, $y(0) = 1$.

